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Paleomagnetism of the Lower Jurassic Lepá and Osta Arena formations, Argentine Patagonia

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Abstract — A paleomagnetic study in Upper Pliensbachian–Lower Toarcian rocks of Extra-Andinian Patagonia was carried out. Characteristic magnetizations isolated from three sections pass the tilt test. The resulting Early Jurassic paleomagnetic pole (PP) (Long. = 129.4°E, Lat. = 75.5°S, $A_{95} = 6.8°$, N = 13, K = 38.7, R = 12.7), together with other reliable Jurassic PPs of cratonal South America defines a hitherto undocumented Jurassic track of apparent polar wander path (APWP). This track suggests recurrent movement of South America with high velocity during the Lower Jurassic. A comparison among the Jurassic PPs of Gondwana continents suggests that APWP can account for a great part of the differences observed between northwestern and southern African Jurassic paleopoles. © 1998 Elsevier Science Ltd. All rights reserved

Resumen — Se realizó un estudio paleomagnético en litologías del Pliensbaquiano Superior–Toarciano Inferior de Patagonia Extraandina. Las magnetizaciones características aisladas en tres secciones, pasan la prueba de estructura. Se obtuvo un polo paleomagnético (Long. = 129.37°E, Lat. = 75.51°S, $A_{95} = 6.8°$, N = 13, K = 38.66, R = 12.7) que junto con otros paleopolos confiables del Jurásico de América del Sur, define una sección antes no documentada de la curva de desplazamiento polar aparente (CDPA) de este continente. La misma sugiere un movimiento recurrente de América del Sur con altas velocidades durante el Jurásico Inferior. Una comparación entre paleopolos de continentes de Gondwana, sugiere que la CDPA puede en gran medida explicar las diferencias observadas entre paleopolos del noroeste y del sur de Africa. © 1998 Elsevier Science Ltd. All rights reserved

INTRODUCTION

Apparent polar wander (APW) paths record the paleolatitude and azimuthal orientation of plates within a hypothetical axial geocentric dipolar Earth Magnetic Field (Irving, 1964; Valencio, 1980; Merril and McElhinny, 1983; Butler, 1992). These paths contain information about the motion of the continental blocks and the motion of the whole (rigid?) outer shell of the planet with respect to the rotation axis, called true polar wander (TPW, e.g. Livermore *et al.*, 1984; Andrews, 1985; Cox and Hart, 1986; Besse and Courtillot, 1991).

After Creer *et al.* (1954) connected a series of paleomagnetic poles (PPs) in a time sequence, producing the first path of APW for Great Britain, the procedure for calculating the paths for different continents, has been progressively modified and improved in response to an increasing amount and quality of paleomagnetic data. A variety of techniques are in use to construct APW paths: (1) The conventional fashion is based on the connection of PPs or their mean that belong to windows spanning of about 25 m.y. This method, developed in earlier times of paleomagnetism, continues in force today (Runcorn, 1956; Van der Voo, 1990, 1993). (2) The "paleomagnetic Euler pole" (PEP) method that rests on the assumption that the paths are caused by plate motion and, citing Euler's theorem, plate motion can be described as rotation in a small circle about some rotation pole (Francheteau and Sclater, 1969; Gordon *et al.*, 1984; May and Butler, 1986; Tarling and Abdeldayem, 1996); (3) The moving window technique averages data using sliding windows centered on multiples of 10 Ma (Irving, 1977 1979; Irving and Irving, 1982). (4) The spherical spline smoothing method, which is connected with the PEP method and with the moving window technique (Jupp and Kent, 1987; Schott *et al.*, 1994), has the advantage of preserving sharp small-scale features such as hairpins that could be smoothed by the other methods. However, all such techniques depend critically upon the available paleomagnetic data bases and the choice of a geologic time scale to refer the selected PPs (Harrison and Lindh, 1982; Van der Voo, 1993).

As is the case for other Jurassic APW paths, that for South America has been the subject of different proposals. Based on the conventional technique, Valencio *et al.* (1983) constructed a Late Paleozoic–Mesozoic APW path for South America and suggested a standstill. Recently, Tarling and

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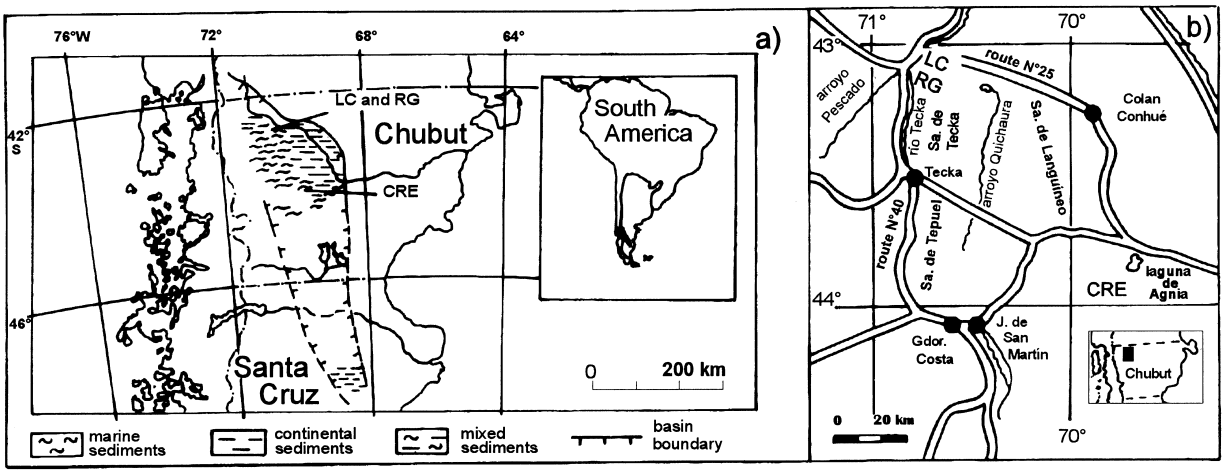


Fig. 1. (a) General location map of the Lower Jurassic Chubut Basin with sampling localities indicated (LC: La Cabaña Section; RG: Río Gualjaina Section; CRE: Cañadón Redondo Epul Section). (b) Sketch map with the main routes of the sampled area; the sampling localities are also indicated.

Abdeldayem (1996) have proposed a small-circle APW path for South America with practically a unidirectional continuous movement throughout the Jurassic, which is in contrast to the proposal of Valencio *et al.* (1983).

On the other hand, the pre-Cretaceous APW path for Africa (the counterpart of South America in West Gondwana) has to be considered with caution. For example, Kosterov and Perrin (1996) observed a difference of about 14° of arc between the Jurassic PPs of northwest and south Africa. They proposed two hypotheses to explain this observation: (1) the ages of the two groups of poles are different; (2) that there has been Early Cretaceous intraplate deformation of Africa. They concluded that there is no paleomagnetic or geological evidence for the first hypothesis, but that the second seems rather plausible from a geological point of view. However, they observed that intraplate deformation models fail to account for the whole difference between northwestern and southern African poles.

In this paper a reliable new late Early Jurassic PP for South America is reported and then the available Jurassic paleomagnetic data are re-evaluated. The selected PPs of South America suggest a high rate of northward motion during the Early Jurassic followed by a slightly slower southward motion during the mid-Jurassic, contrasting with the earlier proposals of a Jurassic standstill and a unidirectional movement of South America during this period. When the South American Jurassic PPs are repositioned in African coordinates using a geologically well-supported Gondwana reconstruction, a comparison with the poles for Africa selected by Kosterov and Perrin (1996) suggests that a difference in age could contribute to the differences between the northwestern and

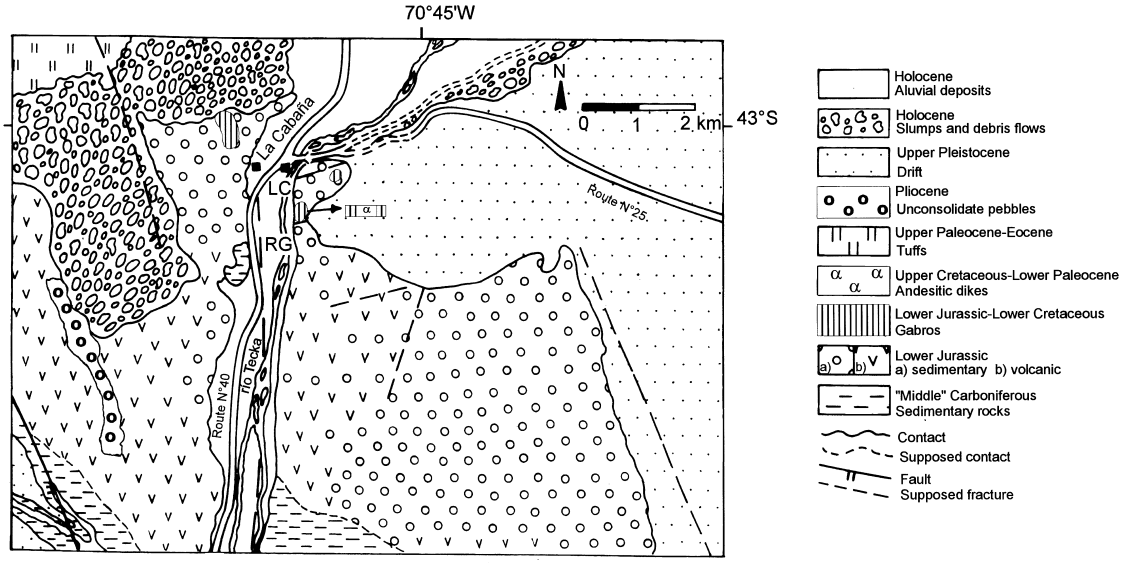


Fig. 2. Geological map of the northern extreme of the Sierra de Tecka with sampling localities LC (La Cabaña Section) and RG (Río Gualjaina Section). Based on Vizán *et al.* (1996).

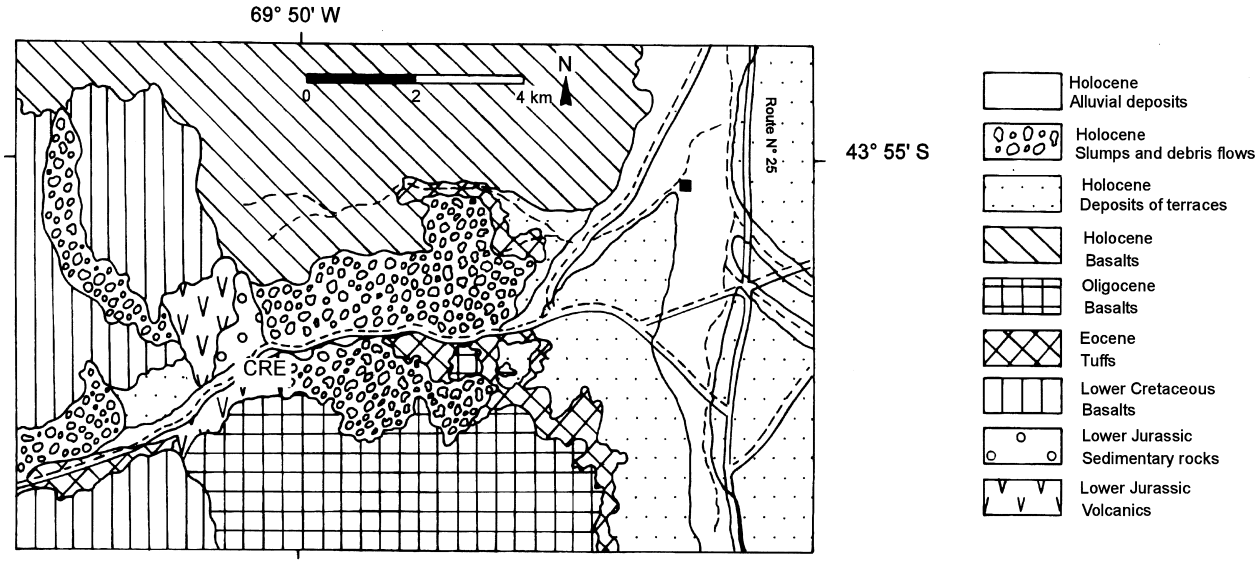


Fig. 3. Geological map of the area of CRE (Cañadón Redondo Epul Section). Based on Nullo (1983).

southern African Jurassic poles. It is speculated that TPW could account mainly for the large movement observed during the Early Jurassic implied by the APW paths of the West Gondwana continents.

GEOLOGICAL SETTING

Paleomagnetic sampling was carried out in three sections of the Early Jurassic Pampa de Agnia basin of Chubut Province (Argentina) (Fig. 1a). The basin has a northwest–southeast trend (Lesta *et al.*, 1980; Riccardi, 1983; González Bonorino, 1990) and the age of its deposits is based on ammonites restricted to the Pliensbachian–Toarcian stages (Blasco *et al.*, 1978; Levy and Blasco, 1981; Von Hillebrant, 1987; Vizán *et al.*, 1996). Three features characterize the different sections of this basin: (1) volcanic and pyroclastic rocks formed part of the sections, (2) the base of the succession is formed by conglomerates with poorly sorted clasts, (3) the upper facies belong to littoral neritic environments (Lesta *et al.*, 1980; Riccardi, 1983; González Bonorino, 1990).

Two of the sampled sections were named "La Cabaña Section" (LC; Lesta *et al.*, 1980) and "Cañadón Redondo Epul Section" (CRE; Nullo, 1983); the other section is very close to LC and here it is informally named "Río Gualjaina Section" (RG). The location of these sections is shown in Fig. 1b. LC and RG belong to Lepá Formation (Rolleri, 1970). These sections are located at the west margin of Gualjaina river, in the northern extreme of the Sierra de Tecka (Fig. 2) where the geology has been studied by Turner (1982) and Vizán *et al.* (1996), among others. The outcropping rocks of the area belong to lower Upper Carboniferous sedimentary rocks with fossils of *Levipústula Zone*; Lower Jurassic volcanic, pyroclastic and epiclastic rocks with *Fanninoceras* sp. (Late Pliensbachian); Lower Jurassic–Lower Cretaceous gabbros; Upper Cretaceous–Lower Paleocene (?) andesitic dikes; Upper Paleocene–Eocene Tuffs; unconsolidated Pliocene deposits with pebbles; Late Pleistocene drift and Holocene alluvial deposits. The rocks of the Lepá Formation were tilted prior to the Cretaceous (Turner, 1982, p. 75).

CRE belongs to the Osta Arena Formation (Herbst, 1966). This section is located in the Pampa de Agnia region where the geology has been analyzed by Nullo (1983) among others. The outcropping rocks of the sampled section (Fig. 3) belong to a Lower Jurassic volcanic, pyroclastic and sedimentary sequence with *Harpoceras falcifer* of Lower Toarcian age (Blasco *et al.*, 1978); Lower Cretaceous basalts; Eocene pyroclastic rocks; Oligocene basalts; Holocene basalts, slumps and debris flow deposits, deposits of terraces and alluvial deposits. According to Nullo (1983), in Cañadón Redondo Epul the tilting occurred before the Eocene.

PALEOMAGNETIC SAMPLING

LC is located at latitude 43°1′ S, longitude 70°44′W. It consists of about 250 m of epiclastic and pyroclastic rocks (Fig. 4). The mean structural attitude of the bedding planes, obtained using traditional Fisher (1953) statistics, is: strike 81°, dip 21°S ($N = 25$, $K = 235.5$). In this and other sections, hand or block samples were oriented with solar and Brunton compasses.

Section LC was divided into six sites, as shown in Fig. 4. These paleomagnetic sites were selected based on the depositional sequences defined by González Bonorino (1990) and González Bonorino and Cesaretti (1990) for the first 150 m of the section. These depositional sequences are separated by minor

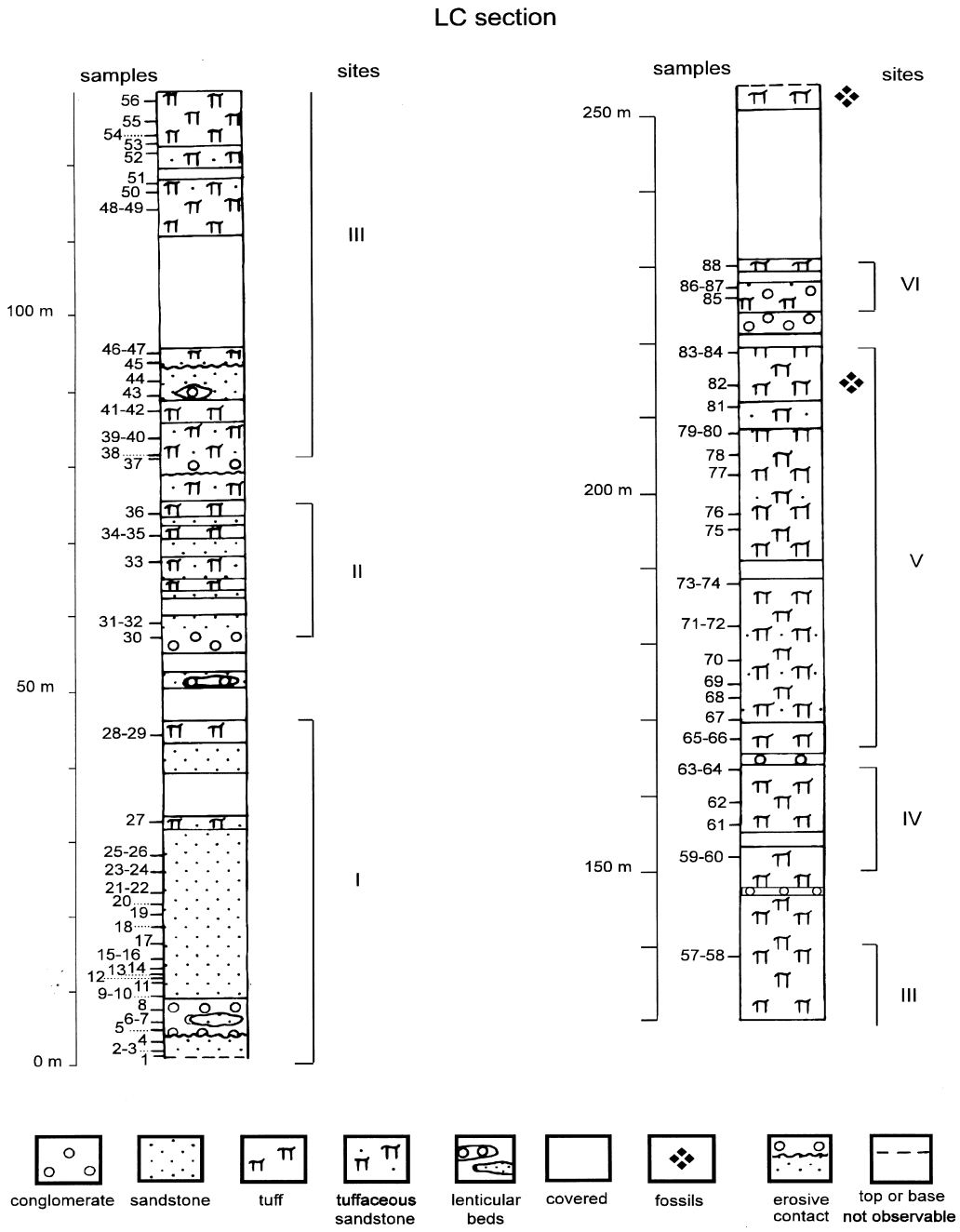


Fig. 4. Stratigraphic column of LC and location of paleomagnetic samples and sites.

disconformities and each of them begins with conglomeratic beds.

RG is located close to LC at latitude 43°1′35″S, longitude 70°44′W. It is 100 m thick and it is mainly composed of andesites, and pyroclastic rocks. The volcanic rocks form the lower and upper parts of the section (Fig. 5). The mean structural attitude of bedding planes is: strike 1°, dip 32°E (N = 3; K = 992).

RG was divided into three paleomagnetic sites according to changes in petrographic characteristics of the section (Fig. 5).

Petrographic analysis of thin and polished sections of rocks from LC and RG reveal the presence of magnetite and titanomagnetite; in some samples recently formed modern iron oxides (goethite?) are associated with carbonates (Fig. 6a). According to González Bonorino and Cesaretti (1990), there is no evidence of strong recrystallization in the tuffs of LC. Also, there is no evidence of metamorphism in the epiclastic rocks of this section (R. Andreis, personal communication).

CRE is located at latitude 43°58′S, longitude 69°51′W. It is 42 m thick and it is composed mainly of red beds including mudstones and sandstones with pyroclastics (Fig. 5). In the upper part of the sections there are conglomerate beds with clasts of quartz. In the sampling area, the Lower Jurassic basalts are strongly deformed and altered by weathering, whereas the sedimentary rocks of the Osta Arena Formation

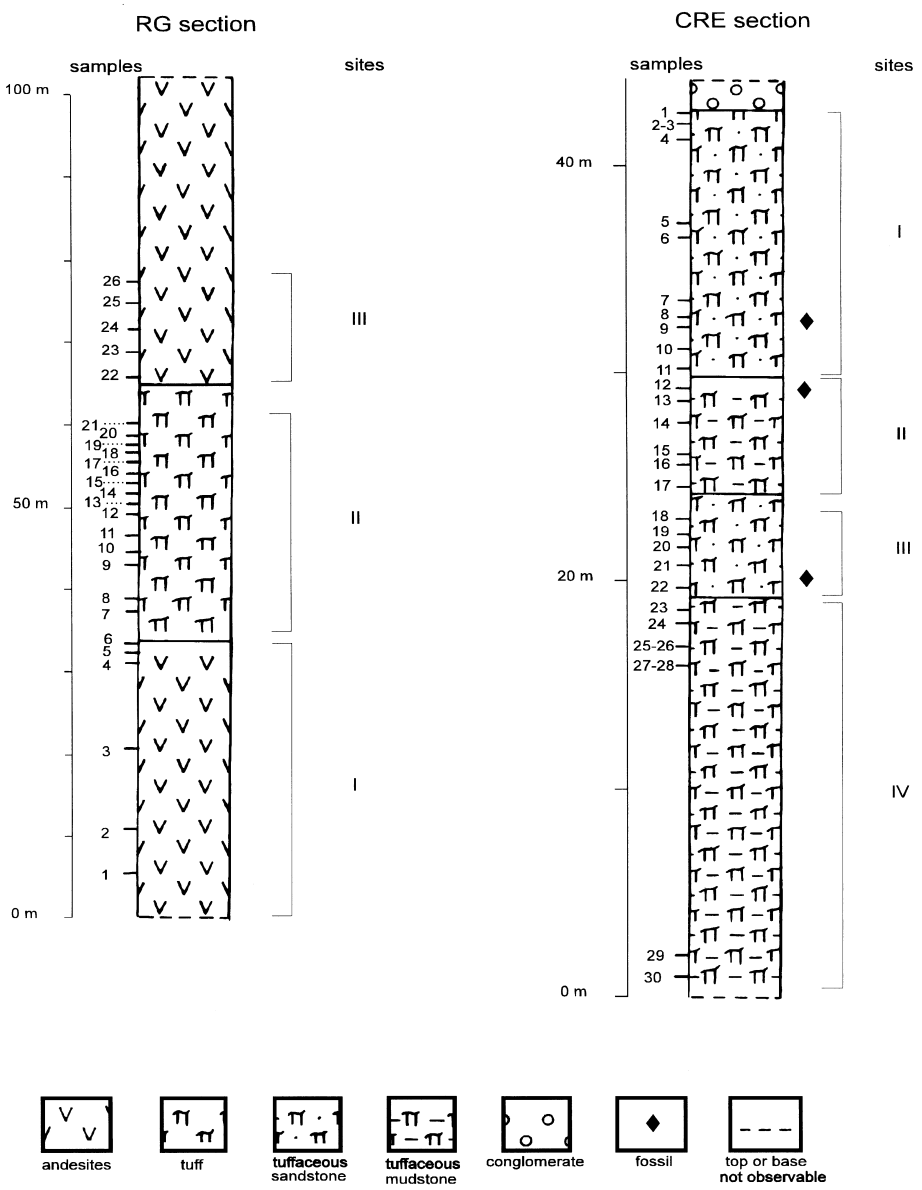


Fig. 5. Stratigraphic columns of RG and CRE and location of paleomagnetic samples and sites.

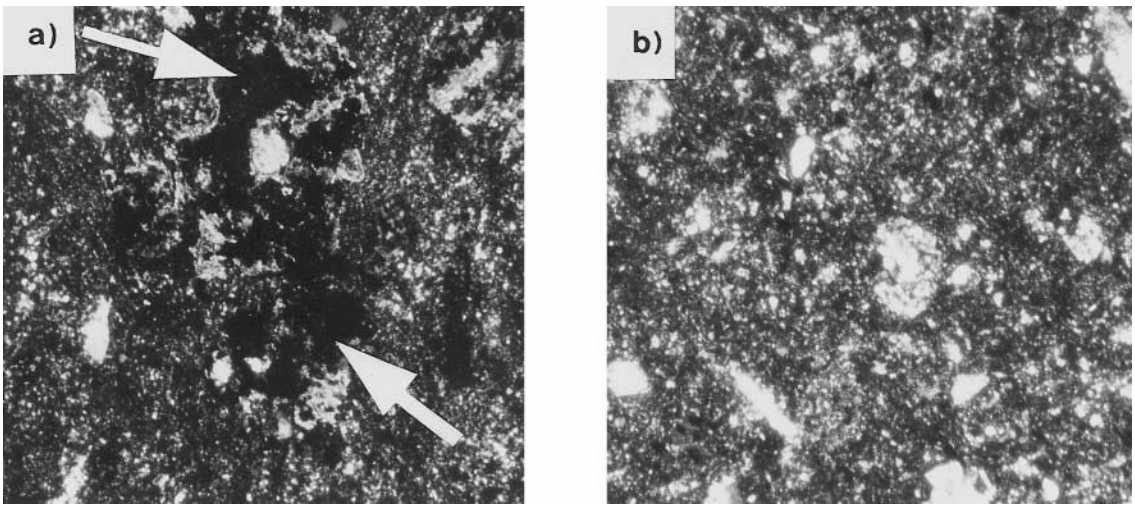


Fig. 6. Photomicrographs of sections of the same sample of LC (×2.5), (a) before, and (b) after applying a chemical treatment with hydrochloric acid. Note that in (a) there are modern iron oxides associated with carbonates (they are pointed out with arrows). (b) These oxides disappeared after the treatment.

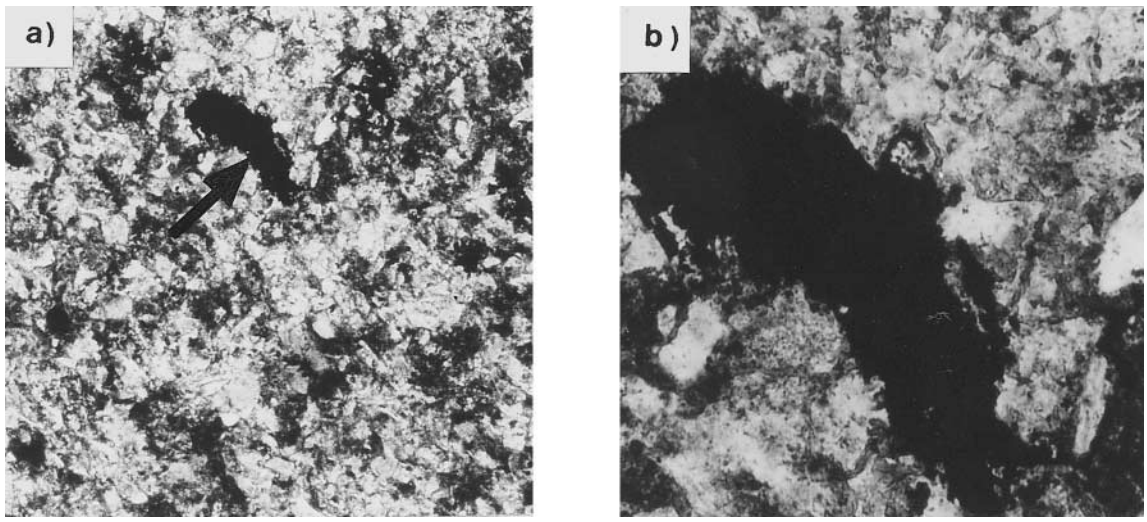


Fig. 7. Photomicrographs of sections of a sample of CRE, (a) general view of the sample with organic material (arrow) (×2.5). (b) detail of this section (×10).

are only tilted and not strongly altered. The mean structural attitude is: strike 11°, dip 14°E (N = 9, K = 114.26). A petrographic analysis previous to and after a chemical treatment with hydrogen peroxide, reveals that several samples contained organic material (Fig. 7a and b), demonstrating that these samples have not undergone strong oxidation.

In CRE four paleomagnetic sites were defined according to changes in petrographic characteristics. Sites I, II, III and IV with, 11, 6, 5, and 8 samples respectively, comprise mainly mudstones or sandstones (Fig. 5).

In summary, all sites include multiple beds and were selected taking geological criteria into account.

EXPERIMENTAL PROCEDURE

General methodology

From the samples from sections LC and RG a total of 142 specimens of 2.5-cm diameter and about 2.5 cm length were cut. From section CRE samples, 32 specimens of similar dimensions were obtained. The natural remanent magnetizations of all specimens before and after alternating field (AF) and thermal demagnetization were determined using fluxgate spinner magnetometers. During thermal cleaning, the bulk susceptibility was checked after each demagnetization step, in order to monitor the occurrence of chemical changes induced by heating.

Lepá formation (LC and RG)

The LC and RG samples showed low values of initial remanence intensity (mean value of 1.45 mA/m). The directions of many specimens at advanced stages of cleaning were corroborated by repeat measurements. Specimens with intensities less than 10^{-1} mA/m were rejected, being at the magnitude of the noise level of the magnetometers.

86 pilot specimens representative of all the sampled lithologies and covering the thickness of both sections were submitted to stepwise AF cleaning, beginning at 7 or 10 mT, and then steps of 5 or 10 mT were performed up to 55–60 mT, depending on the residual intensities. Pilot thermal demagnetization was performed with initial heatings of 70°C (54% of thermal pilot specimens), 100°C (16%) or 200°C (30%). After that, steps of 50° were generally applied up to 450° or 500°C. Steps of 30°, 20° or 10°C were then performed up to 550–580°C, depending on the magnetic behavior and residual intensities of the specimens. The remaining specimens were submitted to at least 5 AF or ther-

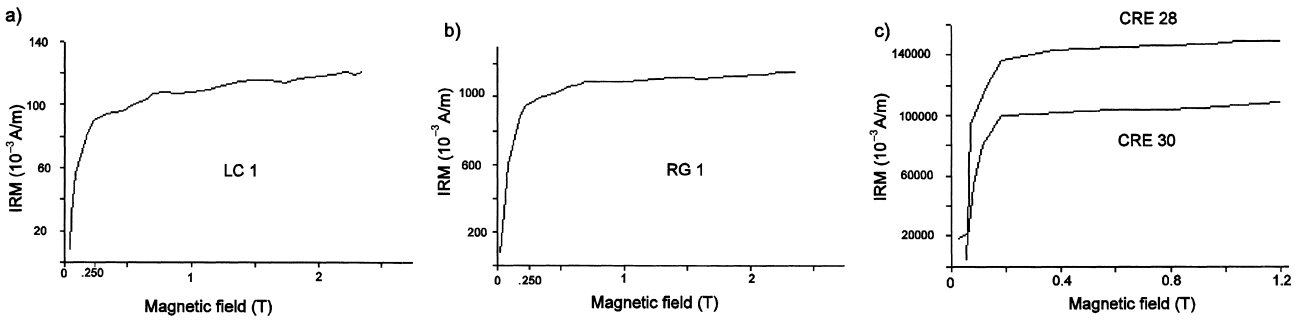


Fig. 8. Isothermal remanent magnetization (IRM) acquisition curves for samples of (a) LC, (b) RG and (c) CRE.

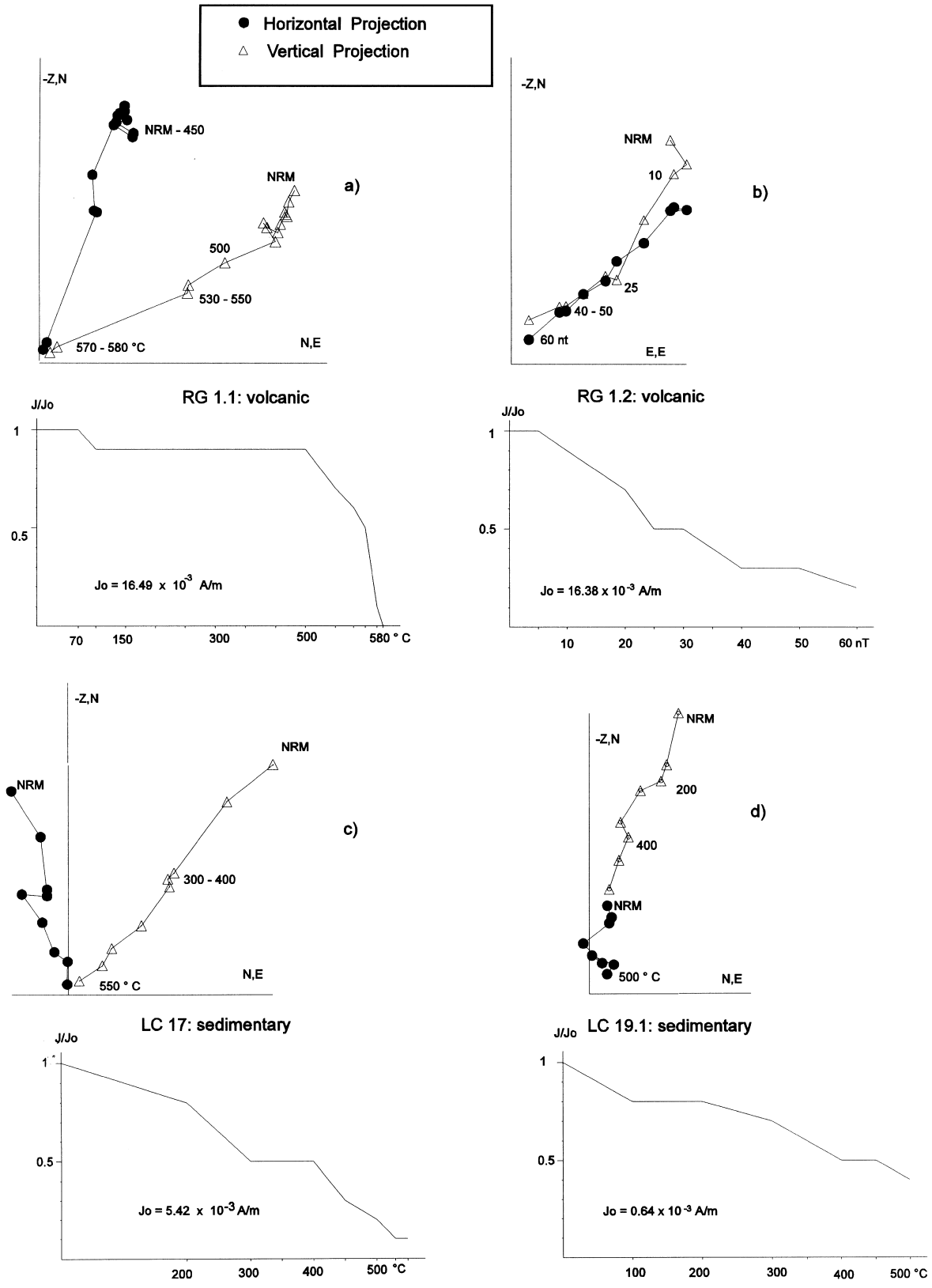


Fig. 9. Magnetic behavior of samples from RG and LC. Open symbols in the Zijderveld diagrams indicate projections on the vertical planes; solid symbols, horizontal planes. Normalized intensity decay after each demagnetization step is shown in the x–y plots. (a) and (b) volcanic specimens. (a) submitted to thermal cleaning. (b) submitted to AF demagnetization. (c) and (d) sedimentary specimens submitted to thermal cleaning. (a), (b) and (c) reliable magnetic behavior (see text for more explanation); (d) moderately reliable magnetic behavior.

Table 1.

La Cabaña (LC): Sierra de Tecka										
	Remanence Directions In Situ				Bedding Correction		Remanence Directions Corrected			
Samples	Dec.	Inc.	k	α_{95}	Strike	Dip	Dec.	Inc.	k	α_{95}
1(2)	53.7	−75.8			76	27	16.7	−59.9		
2–3(2)	80.5	−84.7					5.4	−68.3		
4(2)	13.3	−68.1					3.2	−47.9		
5(3)	77.6	−58.1			72	25	48.7	−51.4		
6–7(2)	328.5	−56.8			81	20	335.8	−36.9		
8(2)	42.3	−36.5					33.6	−22.0		
9–10(2)	12.6	−62.6			84	24	4.3	−42.5		
11(2)	357.9	−56.9					355.6	−36.0		
13(2)	16.2	−68.9					4.5	−49.0		
14(1)	180.3	−63.9					211.9	−83.7		
15–16(3)	317.4	−58.0					328.7	−39.3		
17(1)	346.9	−48.8					348.0	−27.8		
19(2)	12.6	−69.9					2.3	−49.6		
20(1)	274.4	−44.9			83	29	291.6	−36.9		
21–22(2)	353.3	−55.2			80	19	352.6	−34.2		
23–24(2)	266.4	−35.6			88	20	280.0	−31.1		
25–26(2)	70.1	−64.1			85	20	38.0	−54.1		
27(1)	129.8	−56.9			71	18	93.9	−68.4		
28–29(4)	269.1	−33.6			75	25	281.5	−28.3		
30(2)	335.2	−59.4			81	20	340.7	−38.9		
31–32(2)	12.4	−57.8			53	20	5.2	−37.8		
34–35(2)	326.7	−84.2			81	20	345.6	−63.6		
37(2)	12.3	−64.3					3.7	−44.1		
39–40(2)	25.5	−34.1			88	21	20.2	−16.3		
42(1)	289.1	−53.9			79	20	307.6	−40.9		
45(1)	38.7	−80.0					6.5	−61.4		
46–47(2)	49.8	−51.6					33.4	−38.0		
48–49(2)	3.7	−66.6			88	22	358.2	−45.9		
50(1)	42.4	−63.2					22.2	−47.1		
52(2)	304.5	−63.2			83	19	322.8	−46.3		
54–55(2)	29.7	−50.1					19.4	−32.5		
57–58(2)	46.2	−68.0			86	20	21.0	−52.1		
59–60(2)	38.0	−67.7				16.8	−50.4			
62(1)	25.9	−71.1					8.5	−52.0		
63–64(2)	43.6	−63.3			101	21	22.9	−47.4		
65–66(2)	51.9	−53.7			81	20	33.9	−40.5		
71–72(3)	11.9	−55.5			81	18	5.3	−35.4		
73–74(2)	17.7	−69.9			66	22	4.9	−50.0		
75–76(2)	47.2	−73.7					16.4	−57.2		
78(1)	3.0	−67.5					357.7	−46.8		
79–80(2)	53.7	−58.8					32.0	−45.5		
81(2)	35.6	−65.4			60	23	16.9	−47.9		
82(2)	28.6	−38.8			56	21	21.7	−21.4		
85(1)	44.6	−73.1					15.9	−56.2		
86–87(3)	196.9	−87.7					347.8	−71.0		
88(2)	86.7	−46.4			111	22	65.0	−44.5		
Mean	15.44	−70.49	9.97	7			3.53	−50.45	9.97	7
N = 46										
Río Gualjaina (RG): Sierra de Tecka										
1(2)	31.47	−31.53			2	35	7.94	−42.26		
2(1)	58.03	−22.12					43.34	−46.99		
3(1)	3.02	−25.47					348.45	−22.43		
4(1)	311.47	−48.34					298.62	−21.44		
5(1)	29.17	−0.14					25.36	−14.61		
6(1)	35.49	−26.5					15.69	−40.31		
7(1)	25.83	−24.91			1	32	7.92	−33.99		
8(1)	351.21	−68.95					303.93	−49.39		
9(1)	18.11	−43.63					346.03	−44.26		
10(1)	17.6	−5.21					12.44	−13.16		
11(1)	358.26	−37.55					336.85	−29.79		
12(1)	41.04	−28.54					19.58	−44.81		
13(2)	24.75	−34.9					359.24	−41.32		
15(1)	31.48	−36.2					3.81	−45.87		
16(1)	36.57	−13.41					25.28	−29.77		
17(2)	29.21	−30.52					6.73	−40.28		
18(1)	56.08	−36.91					28.45	−58.96		
19(2)	35.2	−41.8					1.2	−51.93		
20(1)	42.04	−9.84					32.65	−29.19		
21(2)	21.21	−56.22					332.98	−53.77		
22(1)	26.41	−40.64					355.25	−46.46		

Bedding corrections were done with the mean structural value (strike 81°, dip 21°S, N = 25, k = 235.5). Number of specimens in parentheses. All remanence directions and bedding corrections values are in degrees. Dec., declination; Inc., inclination; k and α_{95} are Fisherian statistical parameters.

23(1)	23.83	−69.97					310.34	−60.13		
24(1)	38.4	−49.9					352.8	−58.8		
25(1)	332.4	−47.8					312.6	−27.25		
26(1)	54.5	−41.3			0	30	20.9	−61.6		
Mean N = 25	26.53	−37.02	10.5	9.4			358.84	−43.87	10.5	9.4

Bedding corrections were done with the mean structural value (1°/32°, N = 3, k = 992). Other explanation as above.

Cañadón Redondo Epul (CRE): Pampa de Agnia

1(2)	33.7	−36.8			340	17	22.8	−41.0		
2–3(2)	16.8	−61.0			320	20	352.1	−59.4		
4(1)	312.2	−55.3			355	16	304.7	−42.9		
5(1)	24.3	−59.7			20	12	359.8	−60.0		
6(1)	47.0	−22.0			19	12	41.3	−29.7		
7(2)	16.8	−44.4			21	16	3.1	−44.1		
8(1)	45.1	−3.99					43.5	−11.7		
9(1)	27.0	−52.8					7.9	−54.4		
10(1)	10.3	−51.9					353.2	−49.6		
11(1)	24.1	−54.8					3.8	−55.5		
12(1)	25.2	−20.2					19.7	−22.9		
13(1)	356.7	−48.5				20	16	343.1	−43.4	
14(1)	13.8	−46.1					359.5	−45.0		
15(1)	22.5	−40.6					10.1	−41.9		
16(1)	324.6	−39.4				20	16	343.1	−43.4	
17(1)	20.1	−51.7					2.2	−51.7		
18(1)	119.2	−80.6					250.7	−84.2		
19(1)	3.4	−60.0					342.0	−55.5		
20(1)	23.6	−69.6					346.8	−68.1		
21(1)	10.2	−51.2					353.5	−48.9		
22(1)	8.8	−44.4					355.7	−42.2		
23(1)	51.4	−42.4					39.2	−50.4		
24(1)	25.9	−51.0					8.1	−52.5		
25–26(2)	11.2	−52.9					353.4	−50.8		
27–28(2)	1.7	−47.6					347.9	−43.7		
29(1)	5.7	−45.6			352	15	352.3	−42.6		
30(1)	22.3	−50.7					4.8	−51.4		
Mean N = 27	17.99	−50.07	14.57	7.5			1.23	−49.73	14.57	7.5

Bedding Corrections were done with the mean value (11°/14°, N = 9, k = 114). Other explanation as above.

mal demagnetization steps selected for the pilot specimens.

As previously mentioned, some samples of LC are partially dyed by modern oxides of iron (goethite?). Coercivity-spectrum analysis of these samples showed a sharp rise in isothermal remanent magnetization in magnetic fields (H) <300 mT (magnetite or titanomagnetite), followed by increasing IRM acquisition in higher H (Fig. 8a). The specimens from these samples that were submitted to AF cleaning, were previously treated with hydrochloric acid (7 normal) at 70°C up to 170–150 hours. Petrographic analyses before and after this chemical treatment, allow a test of the effectiveness of the method in removing the modern iron oxides to be confirmed (Fig. 6b). Magnetite and titanomagnetite are generally not affected by hydrochloric acid (H. Llambías, personal communication).

For most samples the optimum temperatures and/or alternating fields fell between 400° and 500–550°C and 30 mT and 40 mT respectively. The blocking temperatures of the specimens suggest that the carriers of the magnetization are magnetite or titanomagnetite, which is consistent with their coercivity-spectrum analysis. Figure 8b shows the IRM acquisition of sample RG 1 for which thermal demagnetization is shown in Fig. 9a; this sample shows a sharp rise in IRM in H < 300 mT followed by a slight gradual acquisition of additional IRM in stronger H. The natural remanent magnetism of this sample (Fig. 9a) is removed by thermal demagnetization at 580°C indicating that magnetite is the principal magnetic carrier.

The characteristic remanent magnetization (ChRM) for each specimen was determined by observing the residual directions obtained after several demagnetization steps, controlling their decay to the origin on orthogonal diagrams (Zijderveld 1967), and then applying least-squares line fitting (Kirschvink 1980). Regardless of lithology, 40% of all the specimens treated with AF or thermal demagnetization showed a very reliable behavior, with well defined magnetic directions (MAD value $<5°$) (Fig. 7a, b, and c). 33% of the specimens were moderately reliable with "noisy" magnetic behavior during demagnetization, characterized by increased dispersion of directions obtained after demagnetization steps and less well defined magnetic behavior (MAD values $\geq 10°$ and $\leq 19.5°$) (Fig.9 d).

Specimens showing unstable behavior after submission to both AF and thermal demagnetization were excluded.

Table 1 lists the ChRM directions of samples of each bed: samples 54 and 55 of LC section, although separated (Fig. 4), belong to the same bed. A similar situation occurred for samples 75 and 76. In these cases the directions were averaged. In Table 2 the amount of strata of each site and the corresponding directions are given.

Table 2. Site Mean Directions

				In Situ Tilt Corrected			VGP					
Site	Ns	Ni	Nm	Mdv	Dec.	Inc.	Dec.	Inc.	α_{95}	k	Lat. S	Lon. E
						LC						
I	21	19	17	26.56	13.61	−66.46	3.85	−46.36	10.2	16.08	74.26	121.95
II	5	3	3	13.97	351.76	−68.3	351.41	−47.3	26.9	22	73.8	81.2
III	16	10	9	18.82	24.25	−63.26	11.18	−44.34	12.6	17.53	70.66	140.99
IV	4	3	3	3.62	36.8	−67.52	16.296	−50.09	7.9	242.82	72.2	161.27
V	14	8	8	12.55	31.74	−61.62	16.46	−43.81	9.9	32	67.95	152.16
VI	3	3	3	18.37	76.72	−71.35	32.39	−61.61	37.8	11.7	66.32	208.4
						RG						
I	6	6	5	25.91	31.56	−22.16	15.04	−34.71	21.2	14.03	62.78	141.98
II	15	14	13	17.4	29.5	−31.49	6.204	−41.19	9.8	19	69.77	126.2
III	5	5	5	18.91	24.94	−53.41	338.98	−54.01	22.5	12.47	71.56	40.49
						CRE						
I	10	10	8	19.12	26.66	−48.63	10.24	−50.47	10.7	27.63	74.91	145.88
II	6	6	5	16.15	16.55	−41.93	3.93	−41.74	14.4	29.23	69.82	120.6
III	5	5	4	15.08	10.32	−56.45	350.37	−53.81	13.3	48.64	77.84	69.2
IV	6	6	5	9.6	13.07	−49.98	356.79	−48.48	7	120.9	75.26	99.1

Paleomagnetic pole: N = 13; Lat. = 75.51°S; Lon. = 129.37°E; R = 12.689; $A_{95} = 6.8$; K = 38.66.
Ns: number of sampled strata per site. Ni: number of characteristic directions of each site; Nm: number of directions involved in each mean site direction. The difference between Ni and Nm is the number of directions that were rejected because they have a deviation greater than twice the mean deviation of each site (see text for more information). Mdv: mean deviation considering the characteristic directions of each site (Ni). Dec.: declination (in degrees); Inc.: inclination (in degrees); VGP: virtual geomagnetic poles (in degrees), they are formed by more than one spot reading (see text for more explanation); α_{95} or A_{95} (in degrees) and k or K are Fisher's (1953) statistical parameters. Lat.: latitude in degrees; Lon.: longitude in degrees.

Osta Arena formation (CRE)

The ancient cement of the lithologies of this section is mainly composed of oxides of iron (hematite?), although some sandstone beds have carbonate cement (Nullo, 1983 and this work). Coercivity-spectrum analysis of these samples showed that although they acquired IRM in H < 300–200 mT, they are not completely magnetized at higher fields (Fig. 8c). All the specimens were submitted to thermal cleaning. A first stage of 100°C or 200°C was performed for all specimens; after which steps of 100°C or 150°C up to 400°or 450°C were applied according to the residual intensity and direction. Beyond 450°C, steps of 30°, 20° or 10°C were applied up to the blocking temperatures of the samples. The bulk susceptibility was checked after each stage of demagnetization. The blocking temperatures indicate that carriers of the magnetic remanences belong either to hematite-ilmenite series or the titanomagnetite series (Fig. 10).

The ChRM of CRE specimens were determined using the procedure already described for the LC and RG samples. In about 50% of the specimens, the directions exhibited little change after 300°C (Fig. 10a). In other specimens, it was possible to isolate one direction above stages of 450°C or higher temperatures (Fig. 10b, c and d). MAD values of the ChRM of all CRE specimens are between 1.5° and 10°.

ANALYSIS OF PALEOMAGNETIC DIRECTIONS

To test if the measured magnetizations are primary, a tilt test was performed. The rocks of LC and RG are assigned to consecutive Early Jurassic geological stages (Pliensbachian–Toarcian). The sedimentary rocks of both formations belong to the same eustatic maximum (Hallam, 1982, González Bonorino, 1990). The age of the directions of the ChRM of the three sections was analyzed with respect to the tectonics. A tilt test was carried out for the sample directions and for the site mean directions.

In situ sample directions from the three sections form separate groups that become consistent after structural corrections (Table 1). Figure 11a and 11b show mean sample directions and their respective 95% confidence circles for LC, RG and CRE before and after applying structural corrections. The McElhinny (1964) statistical test was performed and the values of the Fisher statistical parameter kappa before and after structural corrections were respectively 8.53 and 11.14 (number of samples N = 98), which indicates a positive tilt test.

Before applying the tilt test on a site level, the mean direction for each site was obtained using an iterative method of the program MAG88 (Oviedo, 1989) which finds the gravity center of the population and the angular deviation for each direction of a site with respect to the site mean direction calculated by Fisher's statistic. If the angular deviation of a direction was greater than twice the mean angular deviation of all directions of its site, this direction was rejected and the mean direction of the site recalculated (Table 2).

A positive tilt test was also obtained when the site mean directions of LC, RG and CRE were used (Fig. 11c and 11 d). The values of kappa before and after structural correction were 21.8 and 54.4 respectively (number of sites N = 13). McFadden's (1990) tilt test was also performed. As the angles between the *in situ* overall site mean directions and the tilt-

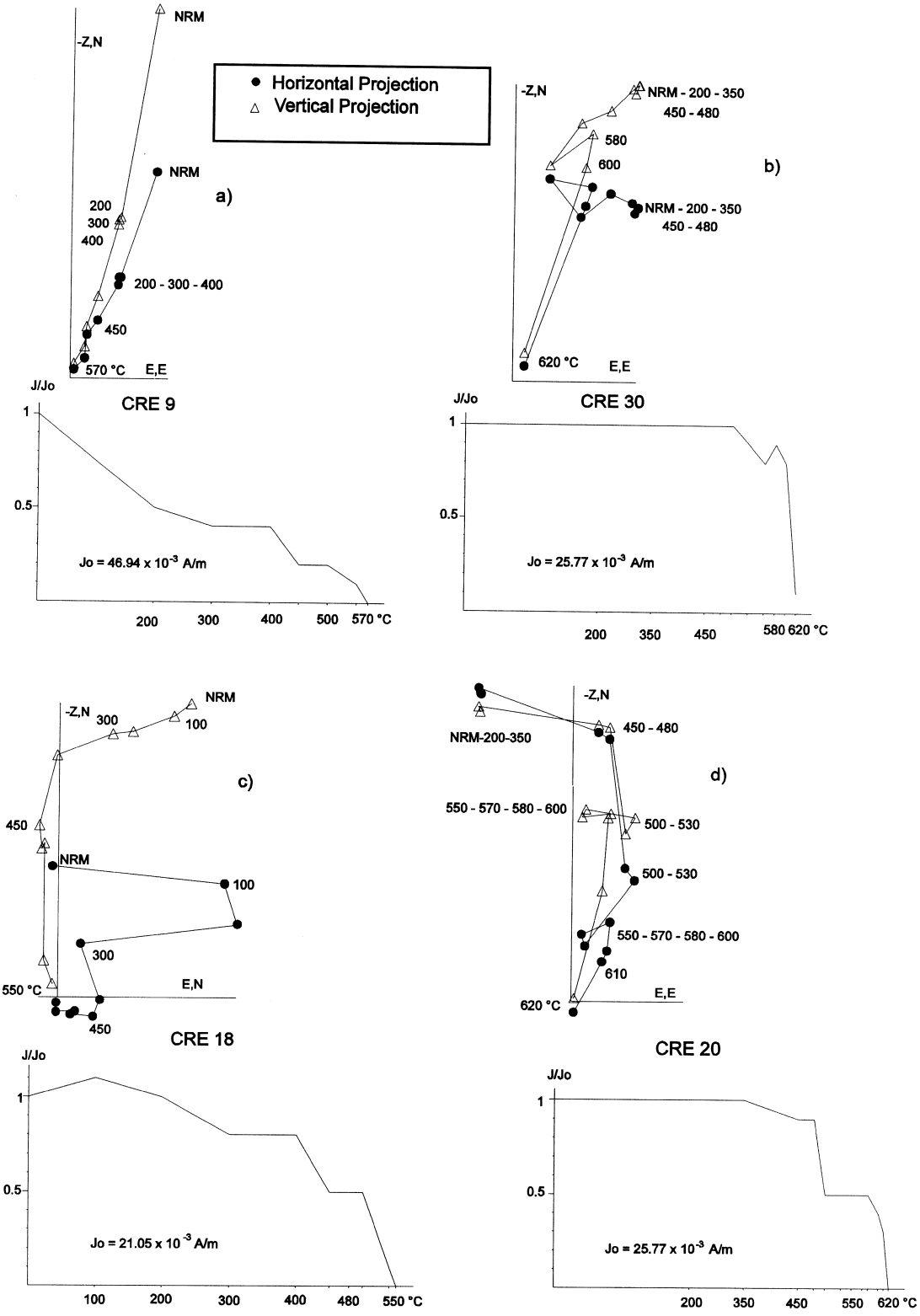


Fig. 10. Magnetic behavior of samples from CRE. Symbols as in Fig. 9. (a) sample for which directions did not experience significant changes above 200°C. (b), (c) and (d) samples with characteristic magnetic remanences isolated above 450°C.

corrected overall site mean directions were small, the second definition of McFadden's test was used. The 95% and 99% critical values were respectively 4.2 and 5.86. A positive tilt test was obtained again. The in situ and tilt-corrected values of SCOS2 were 10.69 and 1.31 respectively.

The age of tilting of the Sierra de Tecka lithologic sequences was pre-Cretaceous (Turner, 1982), so that the age of magnetization should be Jurassic for the ChRM of LC and RG sections. For the ChRM of CRE section a conservative interpretation constrains the age of the magnetization as pre-Eocene, according

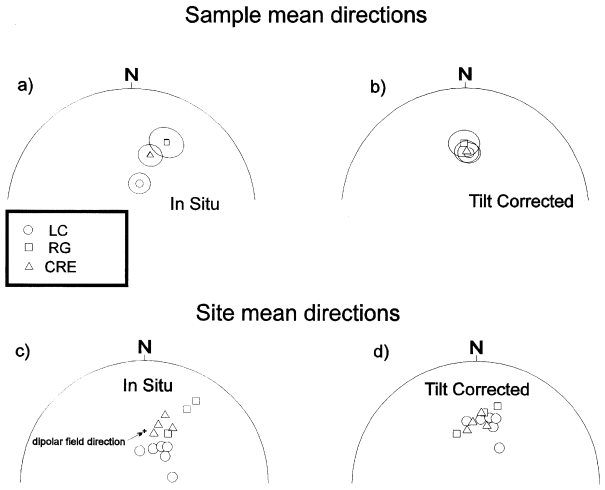


Fig. 11. Averaged sample directions and respective 95% confidence circles (A_{95}): (a) before and (b) after applying the tectonic corrections. Note the overlap of the confidence circles after tilt corrections. (c) and (d) Site mean directions: (c) before tectonic corrections (in situ), (d) after tectonic corrections.

to the geology of Cañadón Redondo Epul (Nullo, 1983).

The characteristic remanence was acquired by different types of rocks (sandstones, tuffs, andesites) and the sampled sections of the Lepá Formation are separated by more than 100 km from the CRE section (Fig. 1b). No evidence of post-Jurassic regional metamorphism has been observed in Extra-Andinian Chubut (Lesta *et al.*, 1980). Indeed, as already mentioned, no evidence of strong recrystalization or metamorphism has been observed in rocks of LC or CRE. Therefore a Lower Jurassic age is assigned for the characteristic remanence, and a primary origin is assumed.

From the site mean directions the corresponding virtual geomagnetic poles (VGPs) were calculated. It is noteworthy that these VGPs are not spot readings of the ancient field. They are all of normal polarity and this is quite remarkable because of the high rate of reversal frequency that has been reported for the late Early Jurassic (i.e., Gradstein *et al.*, 1994; Ogg, 1995; Iglesia Llanos, 1996; Iglesia Llanos *et al.*, 1996). This fact is especially remarkable because section LC is 250 m thick and is late Pliensbachian age according to *Fanninoceras sp.*, as already mentioned. Although there are some normal chrons, reverse polarity prevailed during both Fanninoceras biozones (Iglesia Llanos, 1996, Iglesia Llanos *et al.*, 1996). Rapid sedimentation and consecutive fast acquisition of the remanence during a normal chron could provide a possible explanation. In LC more than 70% of the beds belong to tuffs (Vizán *et al.*, 1996), the absence of sedimentary structures in these tuffs suggests that their deposition was very fast (González Bonorino and Cesaretti, 1990). In fact, the time-span that is involved in the first 150 m of LC could be much less than 4 m.y. (González Bonorino, 1990).

A paleomagnetic pole (PP) based on thirteen site VGPs of LC, RG and CRE was calculated using the method of Oviedo (1989). Its geographic coordinates and statistical parameters are: Long. = 129.4°E, Lat. = 75.5°S, N = 13, $A_{95} = 6.8°$, K = 38.7, R = 12.7.

DISCUSSION

The age of Lepá and Osta Arena formations (LO)PP is assigned to the Late Pliensbachian–Early Toarcian in agreement with the ammonites of LC and CRE. The boundary between the Late Pliensbachian and the Early Toarcian is located at 187 Ma (Ogg,

Table 3. Jurassic paleomagnetic poles from South America with quality factor Q = 3

Pole	Rock unit	Age (Ma)	Latitude (°S)	Longitude (°E)	A_{95}	Q	Reference
VM	Maranhâo Volcanics	175 ± 2 ($^{40}Ar/^{39}Ar$)*	85	263	7	3	Schult & Guerreiro (1979)
CA	Chon Aike Complex	168 ± 2 ($^{87}Rb/^{86}Sr$)**	81.2	207	13.5	4	Recalculated for this study
MF	Marifil Complex	183 ± 2–178 ± 1 $^{87}Rb/^{86}Sr$***	80.5	203.5	8.7	4	Recalculated for this study
LO	Lepá-Osta Arena Fm.	U.Pliensb.–L. Toarc.*****	75.5	129.4	6.8	6	This study
NQ	Jl-Neuquen basin	Pliensb.-Toarc.*****	77	90.7	2	7	Iglesia Llanos (1997)
AT	Anari Tapirapuâ	197 ($^{40}Ar/^{39}Ar$)	65.5	250	4	5	Montes Lauar et al. (1994)
DB	Bolivar Dikes	203 ± 5 Ma ($^{40}K/^{40}$ Ar)	67	245	4	4	MacDonald and Opdyke (1974)

* from Laurenzi (1991) in Raposo and Ernesto (1995).
** from Pankhurst *et al.* (1993)(see text for more information).
*** from Rapela and Pankhurst (1993)(see text for more information).
**** Biostratigraphic age according to ammonites, converted to a geochronological age of 187 Ma using the time scale of Ogg (1995).
***** Biostratigraphic age according to ammonites, converted to a geochronological age of 190 Ma (Iglesia Llanos, 1996, 1997) using the time scale of Ogg (1995).
Pole AT satisfies reliability criteria of Van der Voo (1990) numbers 1,2,3,5 and 7. Pole LO satisfies reliability criteria numbers 1,2,3,4,5 and 7. Pole MF satisfies reliability criteria numbers 1,2,5 and 6. Pole CA satisfies reliability criteria numbers 1,2,5 and 6

Table 4. Coordinates of the poles of great circles

Continent	Number of PPs	Latitude (°S)	Longitude (°E)	λ	coordinates
South America	8	4.5	0.6	0.100	South America
South America	8	23.5	220.6	0.100	Africa
Africa	10	20	257.7	0.043	Africa
India	4	26.5	223.7	0.0062	Africa
Australia	9	15.6	209.7	0.153	Africa
East Antarctica	10	3.6	192	0.058	Africa

$\lambda = 0$ for perfect adjustment of direcctions or paleomagnetic poles with the great circle; $\lambda = 1$ for the worst (Oviedo, 1989).

1995). A preliminary Late Pliensbachian–Early Toarcian PP from the Neuquén Basin (Iglesia Llanos and Vizán, 1996) that passes conglomerate and reversal tests, has practically the same geographic coordinates (Lat = 73°S, Long = 120°E) of the LO PP. A new high quality estimate of this PP, that also passes the tilt test, is listed in Table III (Iglesia Llanos, 1997, thesis in preparation) and is quite close to the LO PP. On the other hand, neither of these paleopoles coincides with the reliable PPs that constitute the Cretaceous track of the South America APW path according to Somoza (1994), nor with a mean Tertiary PP of this continent (Van der Voo, 1993), which discards younger remagnetizations.

Together with LO PP, reliable Jurassic PPs were selected to analyze the track of the South America APW path for this period (Table III). All have "quality" estimates of Q = 3 or greater, using the classification system of Van der Voo (1990, 1993).

In this selection there are newly recalculated PPs for the Chon Aike Complex (CH) and Marifil Complex (MF) (Table 3). The previous version of the CH PP (Vilas, 1974) was based on paleomagnetic data from three localities: Puerto Deseado (Valencio and Vilas, 1970), Bahía de Camarones (Creer *et al.*, 1972) and Estancia La Reconquista (Vilas, 1974). However the volcanic rocks of Bahía de Camarones belong to the Marifil Complex. Three items explain this: (1) the geographic location (44°45′S, 65°65′W North Patagonian Massif, Río Negro Province), (2) the petrologic characteristic (see Rapela and Pankhurst, 1993), (3) the radiometric ages (Rapela and Pankhurst, 1993, among others).

Thus a new PP for Chon Aike Complex has been obtained taking into consideration only the paleomagnetic directions of the localities of Deseado Massif (Santa Cruz Province): Puerto Deseado (47°47′S, 66°W) and Estancia La Reconquista (48°10′S, 68°50′W). These sections belong to the same volcanic complex and have quite similar radiometric ages (de Barrio, 1993; Pankhurst *et al.*, 1993; Alric *et al.*, 1996). The sampled sites of each section comprise volcanic flows that could have erupted within a time interval shorter than the dominant periods of geomagnetic secular variations. In such cases, the procedure of Calderone *et al.* (1990) and Butler *et al.* (1991) was used, in which site-mean directions from stratigraphically adjacent flows were compared using the method of McFadden and Lowes (1981) to determine whether the directions are statistically distinguishable at the 95% confidence level. In this manner, new site-mean directions that belong to "cooling units" in the sense of Butler *et al.* (1991) were obtained. These directions were used for the new Chon Aike PP, for which the geographic coordinates and statistical parameters are: Long. = 207.71°E, Lat. = 81.18°S, N = 13, $A_{95} = 13.5°$, K = 10.33, R = 11.84. Table 4

A new paleopole for the Marifil Complex was also calculated using selected data from Rapalini *et al.* (1993); Mena (1990) and Creer *et al.* (1972). Again the Calderone *et al.* (1990) and Butler *et al.* (1991) procedure was used for the directions of stratigraphically adjacent volcanic flows. The geographic coordinates and statistical parameters for the new Marifil (MF) PP are: Long. 203.51°E, Lat. = 80.46°S, N = 25, $A_{95} = 8.7°$, K = 12.15, R = 23.04.

LO and new versions of CH and MF PPs together with the Anari Tapirapuâ (AT) PP (Montes Lauar *et al.*, 1994) were assessed according to Van der Voo's (1990, 1993) reliability criteria. The values of Q of Diques de Bolivar (DB) and Volcanitas Maranhâo (VM) are those proposed by Van der Voo (1993).

It is worth mentioning that a PP of 202 ± 2 Ma, that passes reversal and contact tests, obtained in Patagonia, would have the same geographic position as that of the AT PP according to Grunow *et al.* (1993); however the coordinates and statistical parameters of this PP are not shown in Grunow *et al.* (1993) paper.

The selected Jurassic PPs of South America (Table III, Fig. 12a) appear to define a loop of the South American APW path (Fig. 12b). This Jurassic tracks were obtained by the spherical spline smoothing method and suggests a recurrent movement of South America as earlier suggested by Vizán (1993) and Iglesia Llanos *et al.* (1996). The APW standstill suggested by Valencio *et al.* (1983) thus is not observed in the Jurassic. Furthermore, if DB and AT PPs are considered as a representative set of data for about 200 Ma, the great circle distance between the mean of these poles and the LO PP is of 33°. These paleomagnetic data indicate a roughly south-north movement bigger than 20 cm. y^{-1} for South America during a time interval of about 13 m.y. that is higher than any plate movements observed today (see DeMets *et al.*, 1994). No strong tectonism has been reported for South America between the Sinemurian and the Pliensbachian (Riccardi, 1983). Therefore

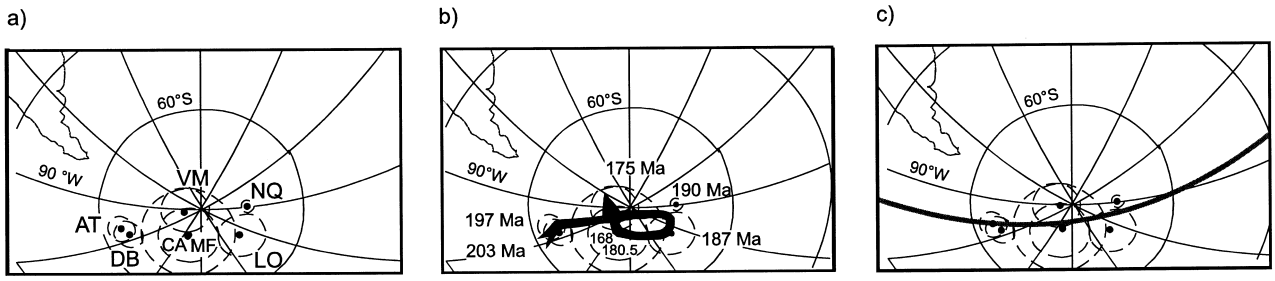


Fig. 12. (a) Selected Jurassic paleomagnetic poles (PPs) of South America (see Table 3) and their 95% confidence circles. (b) Jurassic apparent polar wander path for South America, obtained with the spherical spline method of Jupp and Kent (1987), the ages are those considered for applying the method. (c) Selected Jurassic PPs and the great circle that best contains them. South America in present geographic coordinates.

such a Lower Jurassic movement may be interpreted mainly as true polar wander.

Fine-scale APW movements as that identified in this Jurassic track of South America, would not be recognized if the Jurassic PPs had been separated into windows as broad as 25 m.y. (as those proposed by Van der Voo (1990, 1993) to build an APW path).

The Jurassic APW path here observed, has a quite different trend from the Jurassic segment of the small circle APW path for South America proposed by Tarling and Abdeldayem (1996). To appraise this new trend the South America PPs were compared with Jurassic PPs of other Gondwana continents through continental reconstructions with North Africa in fixed coordinates. The reconstruction poles used are given in the caption of Fig. 13. There are several reconstructions for the fit of West Gondwana, but here the rotation parameters of Nürnberg and Müller (1991) were used. This reconstruction was preferred over others with or without internal deformations (e.g., Rabinowitz and LaBrecque 1979, Pindell and Dewey, 1982), because it has more geological and geodynamical evidence in its favor. The amount of movement affecting the Benue trough is consistent with the crustal stretching of 95 km and left lateral movements of 60 km calculated by Fairhead and Okereke (1990). The implied movement in Chaco-Paraná basin is consistent with the repeated rifting episodes along the trend of this basin (Eyles and Eyles, 1993). Moreover, the Jurassic–Cretaceous basaltic extrusions of the Chaco-Paraná basin are related to strike slip regional faults (Zalán *et al.*, 1991), and the geometry of the Salado and Colorado basins were governed by crustal extension and strike slip faulting (Yrigoyen, 1975; Urien *et al.*, 1981; Uliana *et al.*, 1991). It is noteworthy that the reconstruction of Nürnberg and Müller (1991) did not take into consideration other important geological events in southern South America such as those related to the Golfo San Jorge basin.

PPs used for India, Australia and East Antarctica (with $Q > = 3$) were obtained from the Van der Voo (1993) list, (see caption of Fig. 13). The PPs used for Africa are those recently selected by Kosterov and Perrin (1996). One of the difficulties in comparing PPs of different continents arises from the variable age assignments for the different studies. Another difficulty is that the ages of the PPs even in Van der Voo's recent list, do not permit the definition of a continuous path. Thus, I compare only the distribution of South America Jurassic PPs with other of Gondwana continents without considering any temporal succession. The Jurassic PPs used for the different continents define great circles that were used to make the comparison. Figure 13a, b, c and d and Table IV show that similar great circles can be fitted to the Jurassic PPs of South America and other Gondwana continents, and hence these great circles could reflect similar Jurassic tracks of the APW paths of the Gondwana continents.

The difference between the Jurassic PPs of Southern Africa and North Western Africa selected by Kosterov and Perrin (1996) could be attributed to a difference in ages for the two groups of African poles. Two sets of Jurassic PPs of South America with similar geographic coordinates were averaged using the traditional Fisher statistics. One of these sets was formed by AT, DB and VM (Long. = 53.07°E, Lat. = 68.07°S—north African coordinates—, $A_{95} = 16.9°$, N = 3). The other by LO, NQ, CH and MF (Long. = 92.72°E, Lat. = 57.63°S—north African coordinates—, $A_{95} = 11.6°$, N = 4). These PPs have been compared with the mean PP of southern Africa and the mean PP of northwestern Africa of Kosterov and Perrin (1996) (Fig. 14). The overlap of the South American Jurassic mean PPs with the Jurassic mean PPs of southern and northwestern Africa, suggests that the great circle difference between the African poles could be possibly due to APW path rather than the intraplate deformation in Africa considered by Kosterov and Perrin (1996).

CONCLUSIONS

(1) A paleomagnetic study has been carried out on Late Pliensbachian–Lower Toarcian rocks of the Extra-Andinian Chubut (Patagonia, Argentina). The ChRMs obtained for three sections passes the tilt test. A Lower Jurassic PP

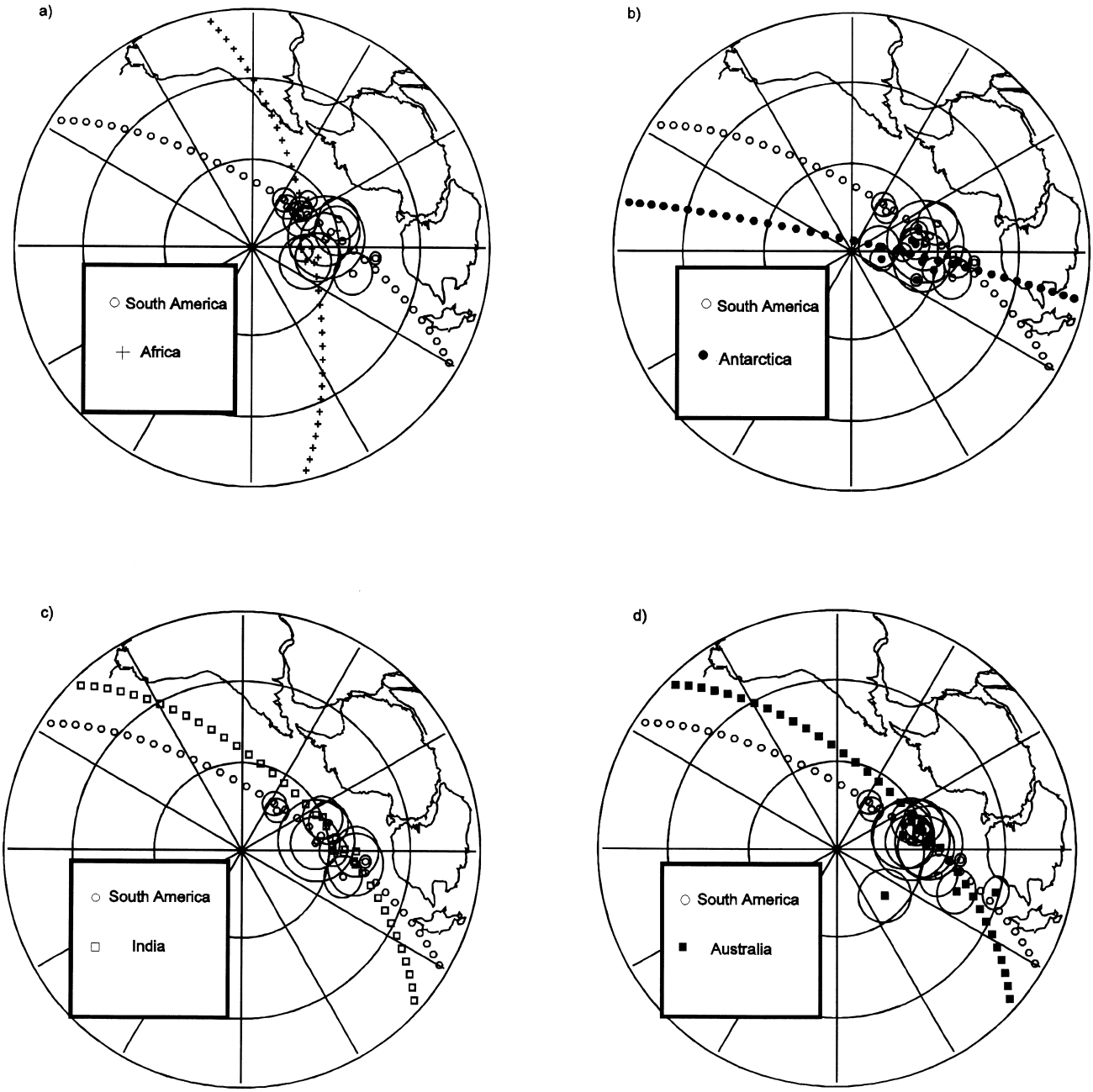


Fig. 13. Comparison of Jurassic PPs with their 95% confidence circles for the Gondwana continents and the great circles that best contain the PPs. Africa in present geographic coordinates. (a) PPs and great circles of South America and Africa. (b) PPs and great circles of South America and East Antarctica. (c) PPs and great circles of South America and India. (d) PPs and great circles of South America and Australia. PPs of Africa are taken from the list of Kosterov and Perrin (1996). Euler pole for South America: Nürnberg and Müller (1991). The other Gondwana PPs were selected from Van der Voo's (1993) list, all of them with Q = 3. PPs of East Antartica: Ferrar Dolerites (FD) Combined, FD McMurdo Sound, FD Wright Valley, FD Mt. Cerberus, Vestfjella Dikes Flows Qu. Maud, Dufec Intrusion Pensacola Mts., Storm Peak Lavas Qu. Alexandra, Mt. Falla Lavas Qu. Alexandra, Vestfjella Flows Dikes Qu. Maud. Euler Pole for East Antartica: Norton and Sclater (1979). PPs of India: Sylhet Traps E. India, Chiltan Ls. Kirthar R. Pak, Loralai Ls. Sanjawi Pakistan, Pachmarhi Red beds Sarpura. Euler Pole for India: Combined Smith and Hallam (1970) and Norton and Sclater (1979). PPs of Australia: Dundas Breccia Pipe, Erskine Park Sill, N.Bondi Volcanic Neck, Hornsby Breccia Sydney Basin, Tasmanian Dolerites West Decl., Luddenham Dike Sydney, Prospect and Other Intr., Sydney Basin Dikes, Garrawilla and Nombi Extr. Euler Pole for Australia: Combined Weissel *et al.* (1977) and Norton and Sclater (1979).

(Long. = 129.37°E, Lat. = 75.51°S, N = 13, $A_{95} = 6.8°$, K = 38.66, R = 12.7) has been obtained using the VGPs corresponding to the ChRM directions.

(2) A new Jurassic APW path of South America has been calculated using this PP together with other PPs considered reliable. This new path suggests a recurrent movement of South

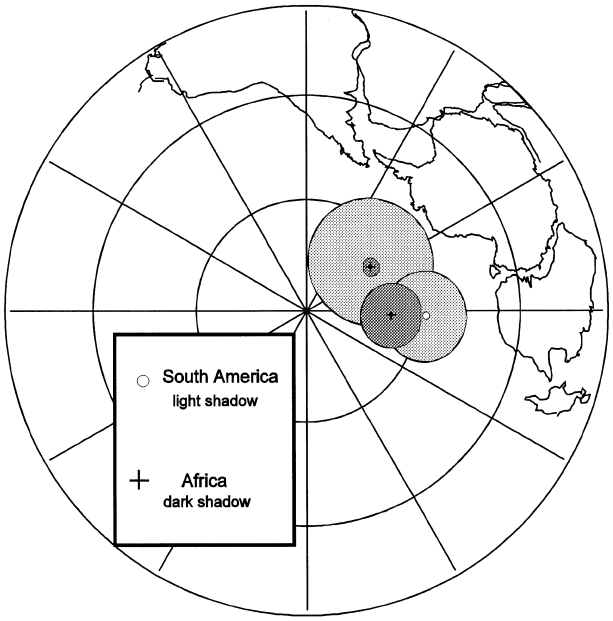


Fig. 14. Mean Jurassic PPs of southern Africa and northern Africa according to Kosterov and Perrin (1996) and mean PPs obtained for the Jurassic of South America. Euler Pole for South America: Nürnberg and Müller (1991). Africa in present coordinates.

America during the Jurassic rather than a standstill as previously suggested.

(3) The high velocity of South America with respect to the paleogeomagnetic axis during the Lower Jurassic could be mainly related to true polar wander.

(4) The trend of the Jurassic APW path of South America is different from that proposed recently by the "small-circle" or PEP method.

(5) A comparison between Jurassic PPs of South America and Africa suggests that the difference of about 14° between mean Jurassic PPs of different parts of Africa could be possibly due to an age difference rather than an internal deformation in this continent.

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